Seminar 4 - ETTI AM2 (PDE)

EDP I și suprafețe de câmp

EDP I

Forma generală:

$$P(x,y,z)\frac{\partial u}{\partial x} + Q(x,y,z)\frac{\partial u}{\partial y} + R(x,y,z)\frac{\partial u}{\partial z} = 0$$

$P, Q, R : \mathbb{R}^3 \to \mathbb{R}$ de clasă (cel puțin) $\mathcal{C}^1$

Not. $\frac{\partial u}{\partial x} = u_x$, $\underline{u(x,y,z) = \text{fc. necunoscută}}$

Pe scurt $Pu_x + Qu_y + Ru_z = 0$.

Se asociază sist. simetric:

$$\rightarrow \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \Rightarrow \text{integrale prime } (I_1, I_2) \quad \begin{cases} I_1 = c_1 \\ I_2 = c_2 \end{cases}$$

! Soluția finală: $u(x,y,z) = \varphi(I_1, I_2)$,

φ funcție: $\mathbb{R}^3 \to \mathbb{R}$ oarecare ($\mathcal{C}^1$)

Ex/24 $(4z-5y)u_x + (5x-3z)u_y + (3y-4x)u_z = 0$

Soluție: Sistemul simetric asociat:

$$\frac{dx}{4z-5y} = \frac{dy}{5x-3z} = \frac{dz}{3y-4x}$$

! Sistemul are sens pentru

$D = \{(x,y,z) \in \mathbb{R}^3 \mid 4z \neq 5y,\ 5x \neq 3z,\ 3y \neq 4x\}$

$\Rightarrow \underline{u : D \to \mathbb{R}}$!

$$\Rightarrow \frac{3dx}{12z-15y} = \frac{4dy}{20x-12z} = \frac{5dz}{15y-20x} = \frac{3dx+4dy+5dz}{0}$$

$$\Leftrightarrow 3dx+4dy+5dz=0 \;\Big|\int \Rightarrow$$

$$\Rightarrow \underline{3x+4y+5z = C_1 = I_1}$$

$$\frac{2xdx}{8xz-10xy} = \frac{2ydy}{10xy-6yz} = \frac{2zdz}{6yz-8xz} =$$

$$= \frac{2xdx+2ydy+2zdz}{0} \Leftrightarrow$$

$$2xdx+2ydy+2zdz=0 \;\Big|\int$$

$$\underline{x^2+y^2+z^2 = C_2 = I_2}$$

! Independența $I_1, I_2 \Leftrightarrow \operatorname{rang}\underbrace{\begin{pmatrix} I_{1x} & I_{2x} \\ I_{1y} & I_{2y} \\ I_{1z} & I_{2z} \end{pmatrix}}_{A} = \max.$

$$A = \begin{pmatrix} 3 & 2x \\ 4 & 2y \\ 5 & 2z \end{pmatrix} \Rightarrow \operatorname{rang} A = 2 = \text{maxim} \Rightarrow$$

$\Rightarrow I_1, I_2$ independente

Soluția finală

$$u(x,y,z) = \varphi(3x+4y+5z,\ x^2+y^2+z^2),$$

$$\varphi \in \mathcal{C}^1(D).$$

Suprafete de câmp

Ex/25 $\vec{V} = xy^2\vec{i} + x^2y\vec{j} + z(x^2+y^2)\vec{k}$

Soluție: Se asociază un OEDPI:

$$xy^2u_x + x^2yu_y + z(x^2+y^2)u_z = 0$$

$\Rightarrow$ sist. simetric asociat:

$$\frac{dx}{xy^2} = \frac{dy}{x^2y} = \frac{dz}{z(x^2+y^2)}$$

$$D = \{(x,y,z) \in \mathbb{R}^3 \mid x\cdot y\cdot z \neq 0\}$$

$$\Rightarrow \underbrace{\frac{y\,dx}{xy^3} = \frac{x\,dy}{x^3y}}_{+} = \frac{dz}{z(x^2+y^2)}$$

$$\frac{y\,dx + x\,dy}{xy\cancel{(x^2+y^2)}} = \frac{dz}{z\cancel{(x^2+y^2)}}$$

$$\Rightarrow \frac{d(xy)}{xy} = \frac{dz}{z} \;\Big|\; S \Rightarrow$$

$$\ln(xy) = \ln z + \ln c_1 \Rightarrow xy = c_1 z$$

$$\Rightarrow c_1 = \frac{xy}{z} = I_1.$$

$$\underbrace{\frac{dx}{xy^2} = \frac{dy}{x^2y}}_{:xy} \left[= \frac{dz}{z(x^2+y^2)}\right]$$

$$\frac{dx}{y} = \frac{dy}{x} \Leftrightarrow x\,dx = y\,dy \;\Big|\int$$

$$\frac{x^2}{2} = \frac{y^2}{2} + c_2$$

$$\Rightarrow x^2 - y^2 = c_2 = I_2.$$

$$A = \begin{pmatrix} y/z & 2x \\ x/z & -2y \\ -\frac{xy}{z^2} & 0 \end{pmatrix}; \operatorname{rang} A = 2 = \max \Rightarrow$$

$\Rightarrow I_1, I_2$ independente

$\Rightarrow u(x,y,z) = \varphi\left(\frac{xy}{z}, x^2-y^2\right)$ pe domeniul D.

Ex/25 $\vec{V} = yz\vec{i} + xz\vec{j} + xy\vec{k}$, $\begin{cases} x^2+y^2=4 \\ y=0 \end{cases}$

Semnalați care este pin γ:

Asociem sist. simetric:

$$\frac{dx}{yz} = \frac{dy}{xz} = \frac{dz}{xy}$$

$$D = \{(x,y,z) \in \mathbb{R}^3 \mid x \cdot y \cdot z \neq 0\}$$

$$\Rightarrow \frac{x\,dx}{xyz} = \frac{y\,dy}{xyz} = \frac{z\,dz}{xyz} \Rightarrow$$

$$\Rightarrow \underbrace{x\,dx = y\,dy}_{x^2-y^2=c_1} \quad \underbrace{x\,dx = z\,dz}_{x^2-z^2=c_2}$$

$$I_1 \qquad I_2$$

$$A = \begin{pmatrix} 2x & 2x \\ -2y & 0 \\ 0 & -2z \end{pmatrix} \quad \operatorname{rang} A = 2 = \max.$$

$$u(x,y,z) = \varphi(x^2-y^2,\ x^2-z^2)$$

<u>Suprafața cerută:</u>

$$I_1, I_2 \Big\} \begin{cases} x^2-y^2 = c_1 \\ x^2-z^2 = c_2 \end{cases} \Big| \cdot 2 \quad \text{compatibil}$$

$$\gamma \begin{cases} x^2+z^2 = 4 \\ y = 0 \end{cases} \Rightarrow 2c_1 - c_2 = x^2+z^2 = 4$$

$$\Rightarrow 2c_1 - c_2 = 4 \Leftrightarrow$$

$$2(x^2-y^2) - (x^2-z^2) = 4 \Leftrightarrow$$

$$\frac{x^2}{4} - \frac{y^2}{2} + \frac{z^2}{4} = 1 \quad \text{hiperboloid cu o pânză.}$$

EDP cvasiliniare

EDP I $u = u(x,y,z) = ?$

EDP cv. $u = u(x,y,\underline{z(x,y)}) = ?$

Exemplu: $(z-y)^2 \frac{\partial z}{\partial x} + xz \frac{\partial z}{\partial y} = xy \left(\frac{\partial z}{\partial z}\right)$

1.26

$$z = z(x,y) = ?$$

Soluția $\exists\, \underline{u(x,y,z) = 0}$ soluție implicită (teorie)

$$u(x,y,z(x,y))$$

$$0 = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial z} \cdot \frac{\partial z}{\partial x} = u_x + u_z \cdot \underline{z_x}$$

$$\Rightarrow z_x = \frac{-u_x}{u_z} \qquad z_y = \frac{-u_y}{u_z}$$

$$0 = \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} \cdot \frac{\partial z}{\partial y} = u_y + u_z \cdot z_y$$

$$(z-y)^2 \cdot \left(-\frac{u_x}{u_z}\right) + xz\left(-\frac{u_y}{u_z}\right) = xy \quad \Big| \cdot (-u_z)$$

$$(z-y)^2 \cdot u_x + xz\, u_y = -xy\, u_z \Leftrightarrow$$

$$(z-y)^2 \cdot u_x + xz\, u_y + xy\, u_z = 0 \quad \underline{\text{EDP I}}$$

$$\Rightarrow \frac{dx}{(z-y)^2} = \underbrace{\frac{dy}{xz} = \frac{dz}{xy}}_{:x}$$

$$\Rightarrow \frac{dy}{z} = \frac{dz}{y} \Leftrightarrow y\,dy = z\,dz \Rightarrow \underline{\underline{y^2 - z^2 = C_1}} \quad I_1$$

$$\Rightarrow \ldots: \underbrace{x\,dx + (y-z)\,dy + (z-y)\,dz}_{0} \Rightarrow$$

$$\Rightarrow \quad x\,dx + y\,dy + z\,dz - \overset{dy\cdot z + dz\cdot y}{d(yz)} = 0$$

$$\Rightarrow \quad x^2 + y^2 + z^2 - 2yz = C_2$$

$$x^2 + (y-z)^2 = C_2 = I_2$$

Soluția finală:

$$\underline{u(x,y,z) = \varphi\left(y^2 - z^2,\ x^2 + (y-z)^2\right) = 0}$$

I_1 și I_2 independente, deoarece

$$\operatorname{rang}\begin{pmatrix} 0 & 2x \\ 2y & 2(y-z) \\ -2z & 2(y-z) \end{pmatrix} = 2 = \max$$